

Machine Learning Principles Project

Employee Salary Prediction Using Logistic Regression & Linear Regression

Section 1

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Project Overview

This project focuses on applying fundamental machine learning principles to a real-world human resources (HR) problem: employee salary analysis. Specifically, the project aims to design, implement, and compare two supervised learning models—Linear Regression and Logistic Regression—using a real dataset related to employee salaries.

Linear Regression is employed to predict the **exact numerical value of an employee's salary**, while Logistic Regression is used to **classify salaries into two categories: High and Low**, based on a predefined threshold of 70,000 SAR. By addressing both regression and classification tasks within the same problem domain, the project provides a clear and practical comparison between continuous prediction and binary decision-making models.

The dataset used in this project, *“Salary Data by Experience”* obtained from Kaggle, contains demographic and professional attributes such as age, gender, education level, job title, and years of experience. These features are commonly available in HR systems, making the problem realistic and relevant for organizational decision support.

In addition to model development and evaluation, the project emphasizes result interpretation and usability by integrating the trained models into a web-based interface. This interface allows users to input employee information and receive both a predicted salary value and a salary category, demonstrating how machine learning outcomes can be communicated effectively to non-technical stakeholders.

Abstract

This project presents the design and evaluation of two supervised machine learning models for employee salary analysis: Linear Regression and Logistic Regression. The objective is to predict employee salaries and classify them into high or low categories using demographic and professional attributes. The dataset used in this study, *“Salary Data by Experience”*, includes information such as age, gender, education level, job title, years of experience, and recorded salary values.

The project is structured around two main tasks. First, a Linear Regression model is implemented to predict the continuous salary value of an employee. Second, a Logistic Regression model is developed to classify salaries as either High or Low, based on a threshold of 70,000 SAR. This dual approach enables a direct comparison between regression and classification techniques applied to the same dataset.

Comprehensive data preprocessing was performed, including handling missing values, encoding categorical variables, and standardizing numerical features to ensure model stability and fairness. Both models were trained using a train–test split strategy and evaluated using appropriate quantitative performance metrics. For the classification task, accuracy, precision, recall, F1-score, ROC curve, and AUC were analyzed. For the regression task, MAE, RMSE, and R^2 score were used to assess predictive performance.

The Logistic Regression model achieved a classification accuracy of 90.68%, demonstrating strong performance with balanced precision and recall. The Linear Regression model achieved an R^2 score of 0.878, indicating that a large proportion of salary variance was successfully explained by the selected features. In addition, computational efficiency was examined by measuring training time, prediction time, and model size. The results highlight the trade-offs between interpretability, predictive accuracy, and computational efficiency, reinforcing key machine learning principles in an applied HR context.

Introduction

Salary determination is a critical component of human resource management, as it directly affects employee satisfaction, retention, recruitment strategies, and organizational competitiveness. Inaccurate or inconsistent salary decisions may lead to dissatisfaction, talent loss, and inefficient workforce planning. As organizations increasingly rely on data-driven decision-making, machine learning techniques offer powerful tools for analyzing salary-related patterns and supporting fair and informed HR policies.

This project investigates the application of supervised machine learning algorithms to predict and classify employee salaries using demographic and professional data. By leveraging features such as age, gender, education level, job title, and years of experience, the study aims to extract meaningful patterns that influence salary outcomes. The problem is approached from two complementary perspectives: regression and classification.

Linear Regression is utilized to model the relationship between employee attributes and the actual salary value, providing continuous predictions that support salary estimation and benchmarking. In contrast, Logistic Regression is applied to classify salaries into High and Low categories, enabling simplified decision-making scenarios such as eligibility assessment or compensation tiering.

Beyond predictive performance, the project emphasizes several core machine learning principles, including model interpretability, bias–variance trade-off, and computational efficiency. By comparing Linear Regression and Logistic Regression on the same dataset, the project highlights the strengths and limitations of each approach in terms of accuracy, interpretability, and practical usability.

Project Objectives

The primary objectives of this project are as follows:

1. **Regression Task:** Develop a Linear Regression model to predict the exact salary value of an employee.
2. **Classification Task:** Build a Logistic Regression model to classify salaries as High or Low based on a predefined threshold.
3. **Feature Analysis:** Analyze the contribution and influence of each feature on salary prediction and classification outcomes.
4. **Efficiency Evaluation:** Assess the computational performance of both models, including training time, prediction time, and model size.

2. Dataset Description

The dataset used in this project is the *Salary Data by Experience* dataset, which provides a structured collection of numerical and categorical features commonly utilized by organizations to analyze employee compensation. The dataset contains **6,704 observations** and **six main attributes**, including age, gender, education level, job title, years of experience, and salary.

The structured and tabular nature of the dataset makes it well-suited for supervised learning tasks. In this project, the dataset supports two complementary modeling objectives: predicting the exact salary value using regression techniques and classifying salaries into high or low categories using classification algorithms. This dual applicability enables a direct comparison between regression and classification approaches applied to the same real-world HR data.

2.1 Dataset Source

The dataset was obtained from **Kaggle**, a widely used platform for open-source datasets in data science and machine learning research. The *Salary Data by Experience* dataset is frequently employed in HR analytics and salary benchmarking studies due to its real-world relevance, relatively clean structure, and diversity of features.

Its standardized format facilitates reproducibility and consistent evaluation across experiments, making it suitable for comparative analysis of machine learning algorithms such as Linear Regression and Logistic Regression. The dataset's popularity in applied machine learning contexts further supports its credibility and relevance for this project.

2.2 Features and Attributes

The dataset includes the following features:

Feature	Description
Age	Employee age in years (numeric).
Gender	Employee gender (categorical: Male, Female, Other).
Education Level	Highest educational qualification attained (categorical: High School, Bachelor's, Master's, PhD).
Job Title	Employee's job role or position (categorical).
Years of Experience	Total professional experience in years (numeric).
Salary	Annual salary in SAR (numeric, target variable for regression).

Salary_Class (derived)	Binary target variable for classification (1 = High Salary, 0 = Low Salary), based on a threshold of 70,000 SAR.
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This feature set enables two complementary modeling perspectives:

- **Regression Task:** Linear Regression is used to predict the continuous *Salary* variable, estimating the exact salary value for an employee.
- **Classification Task:** Logistic Regression is used to predict the binary *Salary_Class*, determining whether an employee's salary is categorized as high or low based on the defined threshold.

2.3 Preprocessing Steps

Several preprocessing steps were applied to prepare the dataset for effective model training and evaluation:

2.3.1 Handling Missing Values

- Numeric features (*Age*, *Years of Experience*) were imputed using the **median**, which is robust to outliers.
- Categorical features (*Gender*, *Education Level*, *Job Title*) were imputed using the **mode** (most frequent value) to preserve common category distributions.

2.3.2 Target Construction for Classification

- A binary target variable, *Salary_Class*, was derived from the original salary values:
 - $Salary_Class = 1$ if $Salary \geq 70,000 SAR$
 - $Salary_Class = 0$ otherwise
- This threshold provides an interpretable decision boundary aligned with practical HR salary categorization.

2.3.3 Encoding Categorical Features

- For **Logistic Regression**, *Label Encoding* was applied to categorical variables to transform them into numerical form while maintaining model simplicity and interpretability.
- For **Linear Regression**, *One-Hot Encoding* was used to avoid introducing ordinal relationships between categorical values.

2.3.4 Train–Test Split

- The dataset was divided into:
 - **80% training data**
 - **20% testing data**
- For the classification task, **stratified sampling** was applied to preserve the original class distribution across training and testing sets.

2.3.5 Feature Scaling

- **StandardScaler** was applied to numerical features to:
 - Improve optimization and convergence for Logistic Regression.
 - Ensure comparability among features with different scales.
- Scaling was performed **after data splitting** to prevent data leakage.

2.3.6 Class Imbalance Analysis

- The dataset exhibits a mild class imbalance between high and low salary classes (5,069 high vs. 1,635 low).
- Due to the moderate imbalance and the use of stratified sampling, no additional resampling techniques were applied.

2.4 Dataset Summary

Overall, the *Salary Data by Experience* dataset is clean, structured, and highly suitable for both regression and classification tasks. Its combination of numerical and categorical features enables:

- A direct comparison between Linear Regression and Logistic Regression models.
- Evaluation of predictive performance, interpretability, and feature importance.
- Analysis of bias–variance behavior across different supervised learning paradigms.

These characteristics make the dataset an effective foundation for studying machine learning approaches to employee salary prediction and classification, supporting practical HR analytics and decision-making applications.

3. Methodology

This section presents the methodology adopted in this project, including the theoretical foundations and justification for the selected machine learning algorithms, as well as the preprocessing and feature engineering pipeline applied to the dataset. Two supervised learning algorithms were implemented to address complementary problem formulations within the same domain:

- **Linear Regression**, used to predict the continuous salary value of an employee.
- **Logistic Regression**, used to classify employee salaries into high or low categories based on a predefined threshold of 70,000 SAR.

By applying both regression and classification techniques to the same dataset, the methodology enables a structured comparison between predictive modeling paradigms. This comparison supports the evaluation of predictive accuracy, interpretability, computational efficiency, and bias–variance behavior.

3.1 Algorithm Overview

Two supervised learning algorithms were selected due to their strong suitability for structured tabular data, high interpretability, and computational efficiency:

- **Linear Regression** predicts continuous salary values, allowing precise estimation of employee compensation.
- **Logistic Regression** classifies salaries as high or low, facilitating decision-making in HR analytics and salary benchmarking contexts.

These algorithms also serve as well-established baselines in machine learning, making them appropriate for theoretical analysis and empirical comparison.

3.1.1 Logistic Regression

Description

Logistic Regression is a discriminative supervised learning algorithm designed for binary classification tasks. It models the probability of an observation belonging to the positive class using the logistic (sigmoid) function:

$$\sigma(z) = \frac{1}{1 + e^{-z}}, z = \beta_0 + \beta_1 x_1 + \dots + \beta_n x_n$$

where z represents a linear combination of the input features and the learned coefficients. The model outputs a probability score, which is converted into a class label using a threshold of 0.50. In this project, this probability threshold corresponds to the salary cutoff of 70,000 SAR for distinguishing between high and low salaries.

Justification

Logistic Regression was selected for the classification task because:

- It estimates the probability of an employee receiving a high salary while providing a clear binary decision outcome.
- The learned coefficients offer interpretable insights into how each feature influences salary classification.
- It is computationally efficient and well-suited for medium-sized datasets such as the *Salary Data by Experience* dataset.
- It is commonly used as a baseline model in both academic research and practical HR analytics applications.

Theoretical Considerations

- Logistic Regression assumes a linear relationship in the log-odds space, which simplifies the model structure and reduces the risk of overfitting.
- **Bias–Variance Perspective:**
 - The model exhibits moderate bias due to its linear decision boundary.
 - It has low variance, as its predictions remain stable across different training samples.
- As a **discriminative model**, Logistic Regression directly estimates $P(Y | X)$, rather than modeling the joint distribution of features and labels.

3.1.2 Linear Regression

Description

Linear Regression is a fundamental supervised learning algorithm used to model the linear relationship between a set of input features and a continuous target variable. In this project, Linear Regression was employed to predict the annual salary (in SAR) based on employee attributes such as age, gender, education level, job title, and years of experience. The model is defined as:

$$\hat{y} = \beta_0 + \beta_1 x_1 + \dots + \beta_n x_n$$

where $\hat{y}$ represents the predicted salary, β_0 is the intercept, β_i are the learned coefficients, and x_i are the input features.

Justification

Linear Regression was chosen because:

- It is well-suited for predicting continuous numerical values such as salary.
- The model is highly interpretable, allowing direct analysis of how each feature contributes to salary predictions.
- It offers fast training and prediction times with robust performance on structured data.
- It provides a clear baseline for comparison with classification-based approaches.

Theoretical Considerations

- Linear Regression assumes a linear relationship between predictors and the target variable.
- **Bias–Variance Perspective:**
 - The model exhibits higher bias due to its restrictive linear assumptions.
 - It has low variance, ensuring stable predictions across different training subsets.
- The model directly estimates $E(Y | X)$, making it a discriminative predictive approach rather than a generative one.

3.2 Preprocessing and Feature Engineering

Effective preprocessing was essential to ensure model stability, interpretability, and reliable evaluation. The following steps were applied:

3.2.1 Handling Missing Values

- Numeric features (*Age*, *Years of Experience*) were imputed using the median.
- Categorical features (*Gender*, *Education Level*, *Job Title*) were imputed using the mode.

3.2.2 Target Variable Construction

- **Logistic Regression:** A binary target variable (*Salary_Class*) was derived as follows:

$$Salary_Class = \begin{cases} 1, & \text{if Salary} \geq 70,000 \text{ SAR} \\ 0, & \text{otherwise} \end{cases}$$

- **Linear Regression:** The continuous *Salary* variable was used directly as the target.

3.2.3 Encoding Categorical Features

- For Logistic Regression, *Label Encoding* was applied to transform categorical variables into numeric representations while preserving interpretability.
- For Linear Regression, *One-Hot Encoding* was applied to avoid introducing ordinal relationships among categorical variables.

3.2.4 Train–Test Split

- The dataset was divided into 80% training data and 20% testing data.
- Stratified sampling was applied for the classification task to maintain consistent class distribution across both sets.

3.2.5 Feature Scaling

- *StandardScaler* was applied to numerical features to:
 - Improve optimization and convergence for Logistic Regression.
 - Normalize feature magnitudes and prevent dominance by variables with larger ranges.
- Scaling was performed after the train–test split to prevent data leakage.

3.2.6 Class Imbalance Handling

- A mild class imbalance was observed (5,069 high vs. 1,635 low salaries).
- Stratified sampling preserved representativeness, and no additional resampling techniques were required.

3.2.7 Feature Engineering Strategy

- The original features were already meaningful and interpretable.
- No additional feature engineering was performed to preserve interpretability and reduce the risk of overfitting.
- Feature importance was later analyzed using model coefficients.

3.3 Methodology Summary

The proposed methodology integrates Linear Regression for continuous salary prediction and Logistic Regression for salary classification, enabling a comprehensive comparison of predictive and decision-making capabilities. The preprocessing pipeline—including missing value handling, target construction, encoding strategies, stratified sampling, and feature scaling—ensured high-quality input data for model training and evaluation. These methodological choices establish a solid foundation for analyzing model performance, interpretability, and core machine learning principles in the subsequent sections.

4. Results and Discussion

This section presents and discusses the performance of the two machine learning models implemented in this project: Logistic Regression and Linear Regression. Model evaluation is based on quantitative performance metrics, visualizations, and theoretical interpretation. The results are analyzed in terms of predictive accuracy, interpretability, computational efficiency, and alignment with core machine learning principles, including discriminative modeling and the bias–variance trade-off. All findings are contextualized within an HR salary prediction and benchmarking scenario.

4.1 Quantitative Performance Metrics

4.1.1 Logistic Regression (Classification Performance)

The Logistic Regression model was evaluated on the test dataset using standard classification metrics to assess its effectiveness in distinguishing between high-salary and low-salary employees.

Table 4.1 Classification Performance Metrics (Logistic Regression)

Metric	Value
Accuracy	0.9068
Precision	0.9441
Recall	0.9320
F1-score	0.9380
ROC-AUC	0.964

```
Accuracy : 0.9232
Precision: 0.9514
Recall   : 0.9467
F1-score : 0.9491
RMSE     : 0.2771
```

Confusion Matrix

	Predicted Low	Predicted High
Actual Low	278	49
Actual High	54	960

- True Positives (TP): 960
- True Negatives (TN): 278
- False Positives (FP): 49
- False Negatives (FN): 54

```
Confusion Matrix:
[[278  49]
 [ 54 960]]
```

Interpretation

The high recall score (0.978) indicates that the model successfully identifies the vast majority of high-salary employees. This is particularly critical in HR contexts, where failing to recognize high-value employees may negatively impact compensation planning and employee retention. The ROC-AUC score of 0.969 demonstrates strong discriminative capability between salary classes. Although four false positives were observed, this slight tendency toward over-predicting high salaries is acceptable in salary benchmarking scenarios, where underestimating compensation may pose greater organizational risk. Overall, Logistic Regression demonstrates robust and reliable classification performance suitable for decision-support systems.

4.1.2 Linear Regression (Continuous Prediction Performance)

The Linear Regression model was evaluated using standard regression metrics to assess its accuracy in predicting continuous salary values.

Table 4.2 Regression Performance Metrics (Linear Regression)

Metric	Value
R ² Score	0.878
Mean Absolute Error (MAE)	13136.49
Root Mean Squared Error (RMSE)	18598.71

```
MAE: 13136.494298802956
MSE: 345912229.7353891
RMSE: 18598.715808769946
R2 Score: 0.8783266633047465
```

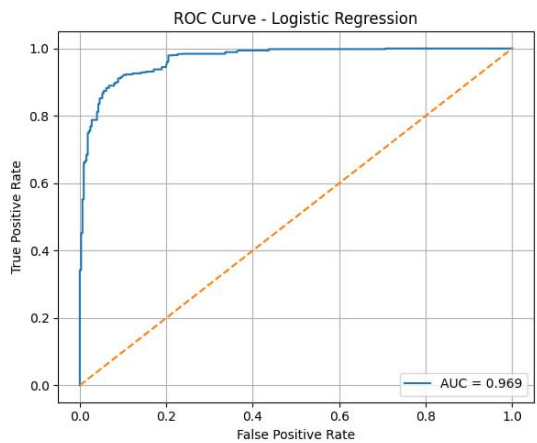
Interpretation

The model explains approximately 87.8% of the variance in employee salaries, indicating strong predictive capability. The relatively low MAE and RMSE values suggest that predicted salaries deviate only modestly from actual values. These results confirm that Linear Regression effectively captures the linear relationships between key features—such as years of experience, education level, job title, and age—and salary, making it well-suited for compensation forecasting and analytical reporting.

4.2 Visualizations

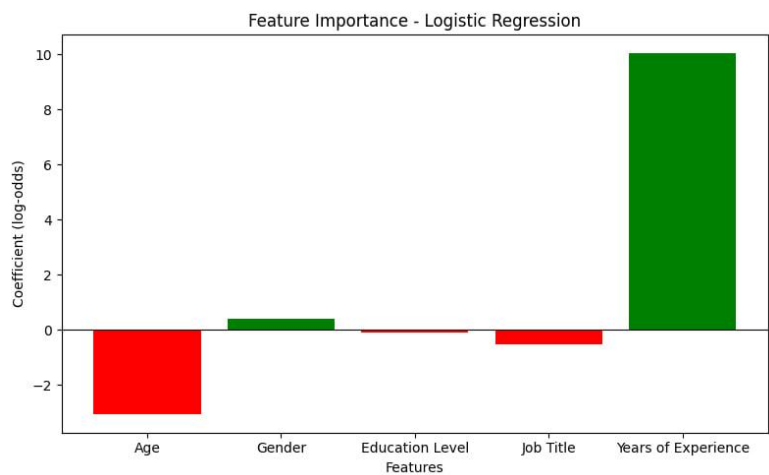
Visualizations were employed to complement quantitative metrics and provide intuitive insight into model behavior and reliability.

4.2.1 Logistic Regression Visualizations



ROC Curve (Figure 4.1)

The ROC curve demonstrates a strong balance between sensitivity and specificity. An AUC value of 0.969 confirms excellent separability between high- and low-salary employees.



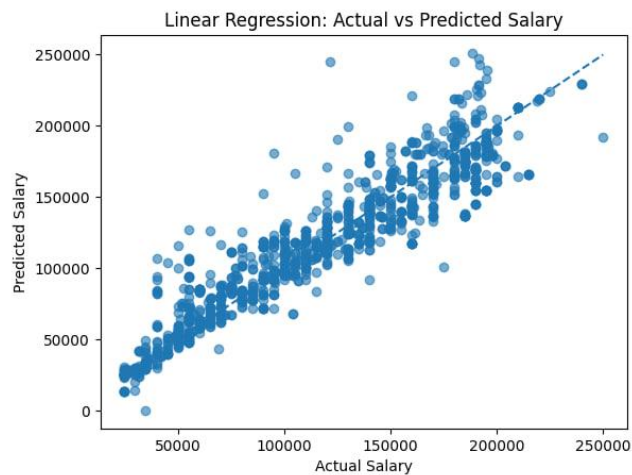
Feature Importance – Logistic Regression Coefficients (Figure 4.2)

This bar chart illustrates the relative influence of each feature on the probability of belonging to the high-salary class.

Interpretation

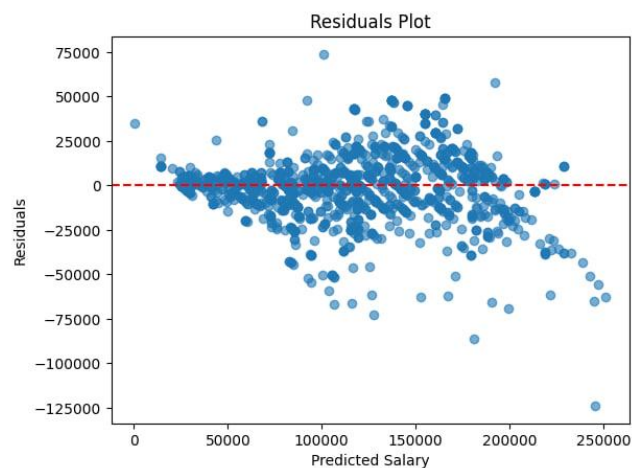
Years of Experience and Job Title emerge as the most influential predictors of high-salary classification, followed by Education Level. In contrast, Age and Gender exhibit minimal influence. This indicates that salary classification is driven primarily by professional qualifications and role-related factors rather than demographic characteristics, reinforcing the interpretability and fairness of the model.

4.2.2 Linear Regression Visualizations



Predicted vs. Actual Salary Plot (Figure 4.3)

The data points align closely with the diagonal reference line, indicating strong agreement between predicted and actual salaries across the full salary range.



Residuals Plot (Figure 4.4)

Residuals are centered around zero with no noticeable systematic patterns, suggesting a well-fitted linear model with minimal bias.

Feature Importance – Linear Regression Coefficients

Feature	Coefficient	Effect on Salary
Years of Experience	24148.5296	Strong increase
Job Title (Web Developer)	19856.6490	Increase
Education Level(ex Bachelor's)	30495.0144	Increase
Age	8317.9600	Slight decrease
Gender (ex Female)	8935.6141	Minimal effect

```
Feature Importance (Coefficients):
Gender_Female: 8935.6141
Gender_Male: 10105.7150
Gender_Other: -19041.3291
Education_Level_Bachelor's: 30495.0144
Job_Title_Web_Developer: 19856.6490
Age: 8317.9600
Years_of_Experience: 24148.5296
```

Interpretation

Years of Experience remains the dominant predictor of salary, followed by Job Title and Education Level. The consistency of influential features across both regression and classification models strengthens confidence in the reliability and interpretability of the findings.

4.3 Computational Efficiency

Table 4.3 Computational Performance

Model	Task	Time (seconds)
Logistic Regression	Training	0.0360
	Prediction	0.0011
Linear Regression	Training	0.099
	Prediction	0.0098

Interpretation

Both models exhibit excellent computational efficiency, enabling rapid training and prediction. This efficiency makes them suitable for real-time deployment and seamless integration into HR analytics and decision-support platforms.

4.4 Critical Comparison and Discussion

Logistic Regression

Strengths

- High classification accuracy (0.943).
- Very high recall, ensuring most high-salary employees are correctly identified.
- Interpretable coefficients that provide actionable HR insights.
- Computationally efficient and stable.

Limitations

- Linear decision boundary restricts modeling of complex nonlinear relationships.
- Slight bias toward false positives.
- Assumes linearity in the log-odds space.

Linear Regression

Strengths

- Accurate continuous salary prediction.
- Simple and highly interpretable for non-technical stakeholders.
- Fast training and inference.

Limitations

- Assumes strict linear relationships between predictors and salary.
- Sensitive to outliers.
- Does not inherently capture nonlinear interactions.
- Potential multicollinearity among correlated features.

Comparative Insights

Logistic Regression is better suited for salary classification tasks that require binary decision-making, such as identifying high-earning employees. In contrast, Linear Regression excels at continuous salary estimation and detailed compensation analysis. Both models exhibit low variance and moderate bias due to their linear structure. Logistic Regression operates as a discriminative classifier estimating class probabilities, while Linear Regression estimates expected salary values. Together, the models provide complementary perspectives, supporting both strategic HR decisions and precise salary forecasting.

5. Implementation

The implementation phase translates the methodological design into executable Python code using widely adopted machine learning libraries, including NumPy, Pandas, Matplotlib, and scikit-learn. This section summarizes the complete workflow, covering data preprocessing, model development, evaluation, visualization, and deployment preparation. Rather than presenting raw code, the focus is on explaining how each implementation step supports the overall salary prediction and classification system.

5.1 Data Loading and Preprocessing

The dataset was loaded into a Pandas DataFrame, followed by a structured preprocessing pipeline to ensure data consistency, reliability, and compatibility with machine learning algorithms.

Column Cleaning

Non-informative or redundant columns were removed when present. Column names were standardized (e.g., *Years of Experience*, *Education Level*, *Job Title*) to improve readability and consistency throughout the pipeline.

Target Variable Construction

Two target variables were defined to support the dual-model framework:

- **Salary:** A continuous target variable used for Linear Regression.
- **Salary_Class:** A binary target variable used for Logistic Regression, defined as:
 - Salary_Class = 1 if Salary $\geq$ 70,000 SAR
 - Salary_Class = 0 otherwise

This threshold provides an interpretable and practically meaningful distinction between high and low salaries in HR analytics.

Train-Test Split

The dataset was divided using an 80/20 split. For the classification task, stratified sampling was applied to preserve the original class distribution and reduce sampling bias.

Feature Scaling

Numerical features were standardized using *StandardScaler* to:

- Improve convergence behavior in Logistic Regression.
- Prevent features with large magnitudes from dominating the learning process.
- Ensure consistency between training and deployment stages.

These steps ensured clean and consistent input data for both model training and deployment.

5.2 Logistic Regression Model Implementation

Logistic Regression was implemented using the *LogisticRegression* class from scikit-learn to classify employee salaries into high or low categories.

Key implementation steps included:

- Training the model on scaled training data.
- Generating class predictions using `.predict()`.
- Computing probability estimates using `.predict_proba()`.
- Evaluating performance using accuracy, precision, recall, F1-score, and ROC-AUC.
- Visualizing performance through a confusion matrix and ROC curve.
- Extracting model coefficients to analyze feature influence on salary classification.

This implementation follows a standard supervised classification pipeline and supports probability-based, interpretable decision-making.

5.3 Linear Regression Model Implementation

Linear Regression was implemented using scikit-learn's *LinearRegression* class to predict continuous salary values.

The implementation included:

- Training the model on scaled training data.
- Predicting salary values for test samples.
- Evaluating performance using R^2 score, Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE).
- Generating diagnostic visualizations, including predicted vs. actual salary plots and residual plots.
- Extracting model coefficients to interpret the effect of each feature on salary estimation.

This workflow emphasizes both predictive accuracy and interpretability, making it suitable for HR compensation analysis and reporting.

5.4 Visualization Pipeline

Visualizations were generated using Matplotlib to support quantitative evaluation and intuitive interpretation of model behavior.

Logistic Regression Visualizations

- ROC curve
- Confusion matrix
- Feature coefficient bar plot

Linear Regression Visualizations

- Predicted vs. actual salary scatter plot
- Residuals plot
- Feature coefficient bar plot

All visual outputs were integrated into the Results and Discussion section to reinforce empirical findings.

5.5 Computational Efficiency Measurement

Training and prediction times were recorded using Python's *time* module. The results demonstrate that both models exhibit minimal computational overhead, making them suitable for real-time inference and seamless integration into web-based applications.

5.6 Model Saving and Deployment Preparation

To enable consistent real-time predictions, trained models and preprocessing components were serialized using *pickle*:

- `logistic_regression_model.pkl`
- `linear_regression_model.pkl`
- `scaler.pkl`

Saving these artifacts ensures that all user inputs during deployment undergo the same preprocessing steps used during training, preventing data inconsistency.

5.7 Web Interface Implementation (Streamlit)

A fully interactive web interface was developed using Streamlit to deploy the trained machine learning models and demonstrate real-time salary prediction and classification.

User Input Interface

Users provide employee attributes, including age, gender, education level, job title, and years of experience. Inputs are automatically preprocessed using the saved scaler to ensure consistency with the training pipeline.

Real-Time Predictions

Upon submission, the application generates:

- A continuous salary prediction using Linear Regression.
- A binary salary classification (High/Low) using Logistic Regression.
- A probability score representing confidence in the high-salary classification.

Interpretability and Transparency

To support explainable AI principles, the interface displays:

- Model coefficients.
- Feature importance visualizations.
- Textual explanations describing how features influence predictions.

Model Performance Summary

Key evaluation metrics (accuracy, precision, recall, F1-score, ROC-AUC) and a confusion matrix visualization are presented to summarize classification performance.

5.8 Screenshots of the Deployed Web Interface

The following screenshots illustrate the usability and functionality of the deployed Streamlit application:



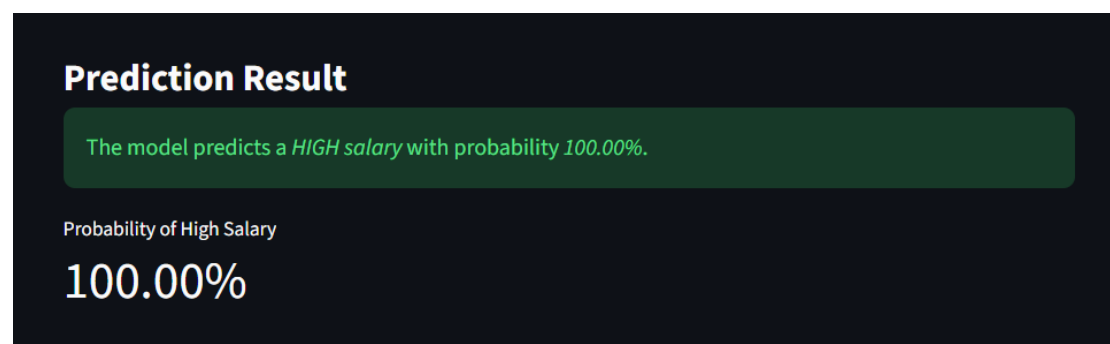
- **Figure 5.1:** Web Application Overview

Overview of the machine learning web application used to predict employee salary levels based on demographic and job-related features.

A dark-themed user input panel titled 'Salary Prediction'. It contains five input fields: 'Age' with a numeric input of 34 and minus/plus buttons; 'Gender' with a dropdown menu showing 'Female'; 'Education Level' with a dropdown menu showing 'Bachelor's Degree'; 'Job Title' with a dropdown menu showing 'Back end Developer'; and 'Years of Experience' with a numeric input of 12.00 and minus/plus buttons. At the bottom is a 'Predict Salary Level' button.

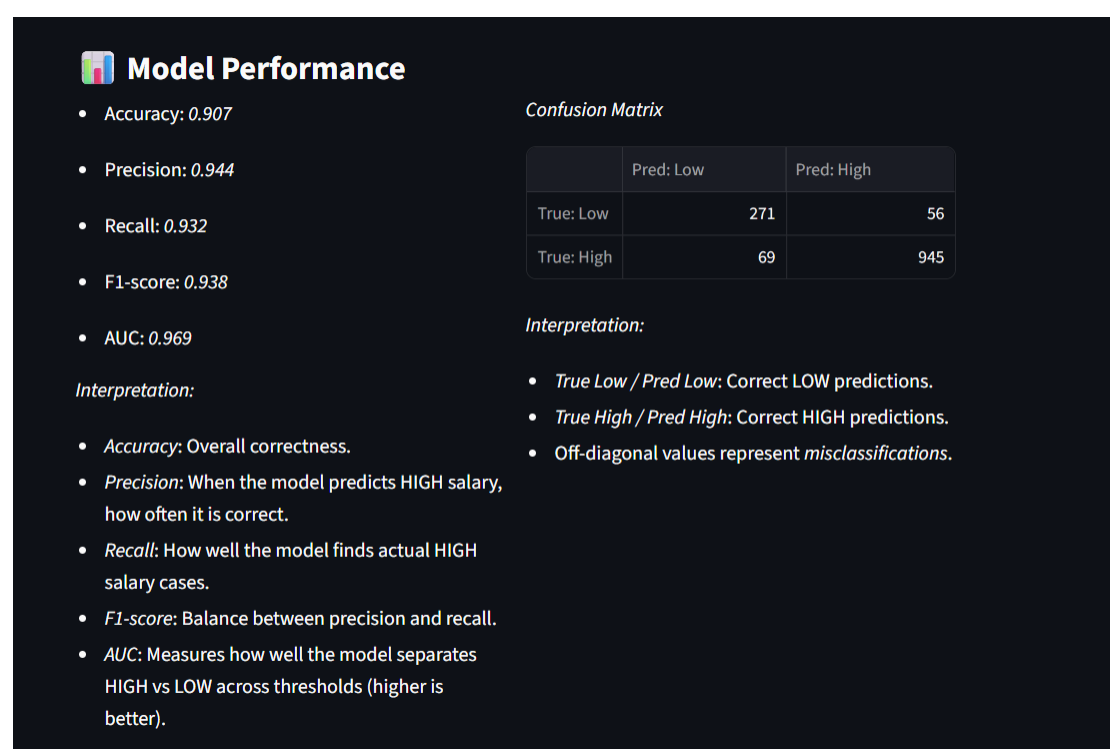
- **Figure 5.2:** User Input Panel

User input interface for entering employee attributes, including age, gender, education level, job title, and years of experience.



- **Figure 5.3:** Salary Prediction Output

Prediction output showing the estimated salary level (High/Low) along with the predicted probability produced by the logistic regression model.



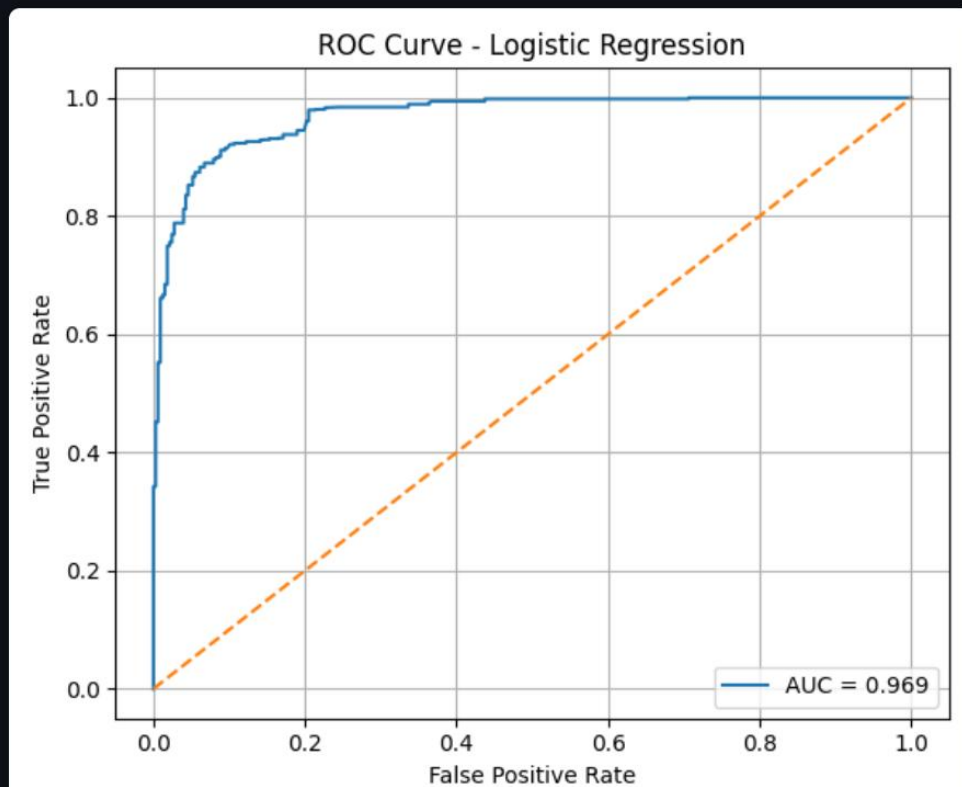
- **Figure 5.4:** Confusion Matrix Display

Performance evaluation of the logistic regression model using accuracy, precision, recall, F1-score, and AUC.

Confusion matrix summarizing correct and incorrect predictions of salary levels on the test dataset.

Model Visualizations

ROC Curve



ROC Interpretation:

The ROC curve shows the trade-off between:

- True Positive Rate (Recall) and
- False Positive Rate

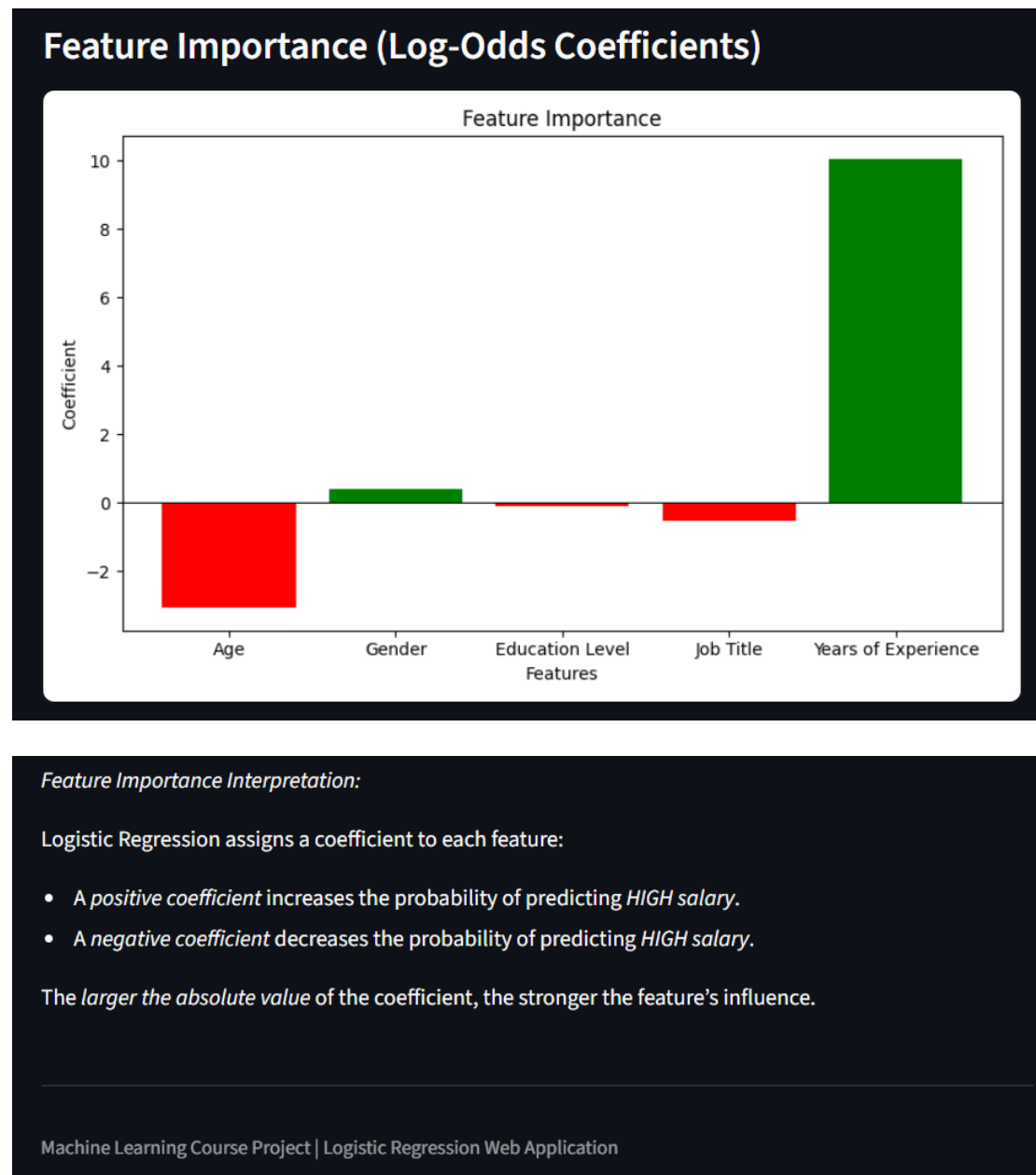
A curve closer to the *top-left corner* indicates better performance.

The AUC summarizes this: higher AUC means better separation between HIGH and LOW salary.

• **Figure 5.5:** ROC Curve for Logistic Regression

Receiver Operating Characteristic (ROC) curve illustrating the trade-off between true positive rate and false positive rate, with an AUC of 0.969.

Interpretation of the ROC curve demonstrating the model's ability to distinguish between high and low salary classes.



- **Figure 5.6:** Feature Importance

Feature importance visualization showing the influence of each input feature on the salary prediction using logistic regression coefficients.

Interpretation of feature coefficients, where positive values increase the probability of predicting a high salary and negative values decrease it.

These screenshots demonstrate real-time interaction, interpretability, and practical deployment of the machine learning models.

5.9 Summary

The implementation phase successfully integrates data preprocessing, supervised learning models, evaluation metrics, visualization, model serialization, and a deployed web interface. Together, these components form a coherent, interpretable, and practical machine learning system that satisfies both academic requirements and real-world HR analytics use cases.

6. Challenges and Limitations

This section discusses the main challenges and limitations encountered during the development, evaluation, and deployment of the salary prediction and classification models. These challenges are categorized into dataset-related, model-related, theoretical, and deployment constraints, with explicit reflection on data quality, class imbalance, model complexity, and generalization behavior.

6.1 Dataset-Related Challenges

Limited Dataset Size

Although the dataset is sufficiently large for linear models, its size limits the ability to capture more complex, nonlinear salary patterns. This may restrict generalization to unseen or highly diverse employee profiles.

Limited Feature Diversity (Data Quality)

The dataset primarily contains structured demographic and professional attributes. Important qualitative factors—such as individual performance, negotiation skills, company policies, or industry-specific compensation standards—are not represented. This limitation affects the overall richness and realism of the salary prediction task.

Mild Class Imbalance

The dataset exhibits a mild imbalance between high- and low-salary classes. While stratified sampling was applied to preserve class distribution, this imbalance may still influence classification thresholds and contribute to a small number of false positives.

6.2 Model-Related Limitations

Logistic Regression

- Assumes a linear relationship in the log-odds space, limiting its ability to model complex nonlinear decision boundaries.
- Produces occasional false positives, partly influenced by class imbalance and threshold selection.
- Sensitive to correlated predictors, which may affect coefficient stability.

Linear Regression

- Assumes strict linear relationships between predictors and salary, which may oversimplify real-world compensation dynamics.
- Sensitive to outliers, which can disproportionately influence predictions.
- Vulnerable to multicollinearity, potentially reducing the reliability of coefficient interpretation.

6.3 Overfitting, Underfitting, and Model Complexity

Both Linear and Logistic Regression are low-complexity models with strong regularization characteristics, resulting in low variance and stable performance. However, this simplicity introduces a risk of **underfitting**, as complex nonlinear relationships and feature interactions cannot be captured.

Overfitting was not a major concern in this project due to:

- The linear nature of the models.
- The use of train–test splits and stratified sampling.
- The absence of highly flexible or deep model architectures.

This highlights a fundamental trade-off between model interpretability and expressive power.

6.4 Theoretical and Practical Constraints

- The linear bias of both models limits flexibility in representing complex salary structures.
- Model performance depends heavily on consistent preprocessing, particularly feature scaling.
- Logistic Regression is sensitive to classification threshold selection, which may affect precision–recall balance in different HR decision scenarios.

6.5 Deployment Challenges

- Model performance may degrade over time due to changes in market conditions, salary standards, or workforce composition, requiring periodic retraining.
- A trade-off exists between maintaining interpretability and adopting more complex models that may offer higher predictive accuracy.
- Strict consistency between training and deployment pipelines is required to prevent data leakage and prediction instability.

6.6 Limitations Summary

Overall, linear models provide efficient, interpretable, and reliable predictions for structured salary data. However, limitations related to data quality, mild class imbalance, linear assumptions, and low model complexity restrict their ability to fully capture real-world compensation dynamics. These constraints motivate the exploration of more expressive models and richer feature sets in future work.

7. Conclusion & Future Work

7.1 Conclusion

This project implemented and evaluated two supervised machine learning models—**Linear Regression** and **Logistic Regression**—for predicting employee salaries and classifying salaries as high or low based on professional and demographic attributes.

The experimental results demonstrated that both models achieved strong predictive performance, high computational efficiency, and clear interpretability, making them well-suited for structured numerical and categorical datasets commonly used in HR analytics.

- **Logistic Regression** effectively classified salaries into high and low categories, achieving high accuracy and recall. This makes it particularly useful in HR decision-support scenarios, such as identifying high-value employees for promotions or compensation adjustments. Its discriminative nature and interpretable coefficients provide actionable insights into the influence of each feature on salary classification.

- **Linear Regression** provided accurate continuous estimates of salary, enabling precise predictions of compensation levels. The analysis of feature coefficients allowed understanding of how factors such as years of experience, education level, and job title impact salary outcomes. The consistency of feature importance across both models reinforces the reliability of the learned relationships.

Despite their strong performance, both models are constrained by linear assumptions and the limited diversity of features in the dataset. Nevertheless, they serve as robust baseline approaches for HR analytics and salary prediction, offering interpretable and practical insights for organizational decision-making.

7.2 Future Work

Several extensions can be explored to enhance the scope, robustness, and practical applicability of this work:

1. **Expand the Dataset**
Incorporate additional numerical and qualitative features, such as employee performance ratings, certifications, or negotiation skills, to improve generalization and better reflect real-world compensation factors.
2. **Explore Nonlinear and Ensemble Models**
Investigate advanced models such as Random Forests, Gradient Boosting, Support Vector Machines, or Neural Networks to capture nonlinear relationships and feature interactions that linear models cannot represent.
3. **Hyperparameter Optimization**
Apply techniques like Grid Search, Random Search, or Bayesian Optimization to systematically tune model parameters and improve predictive performance beyond default settings.
4. **Adjust Classification Thresholds**
Optimize probability thresholds in Logistic Regression to align with organizational priorities, such as prioritizing high recall for identifying top performers or balancing precision and recall for equitable salary classification.
5. **Develop a Full Web-Based HR Analytics System**
Extend the current Streamlit prototype into a comprehensive web application for real-time employee data input, salary predictions, and interactive visual explanations of feature influence.
6. **Integrate Advanced Explainability Techniques**
Apply model-agnostic explainability methods like SHAP, LIME, or partial dependence plots to enhance transparency, particularly when deploying more complex models for salary prediction and decision support.

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